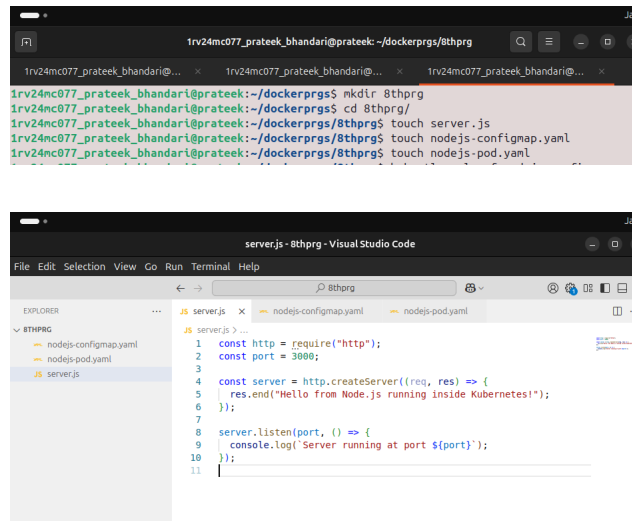


Progam 8

STEP 1: write a node.js app http Server (`server.js`)



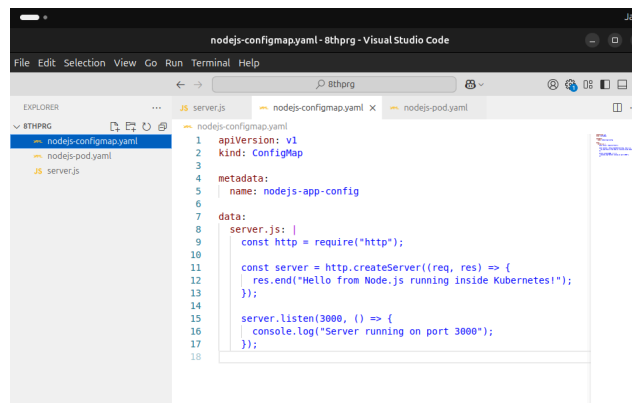
The first screenshot shows a terminal window with the following commands and output:

```
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/8thprg$ mkdir 8thprg
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs$ cd 8thprg/
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/8thprg$ touch server.js
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/8thprg$ touch nodejs-configmap.yaml
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/8thprg$ touch nodejs-pod.yaml
```

The second screenshot shows the Visual Studio Code editor with the `server.js` file open. The code is as follows:

```
1 const http = require("http");
2 const port = 3000;
3
4 const server = http.createServer((req, res) => {
5   res.end("Hello from Node.js running inside Kubernetes!");
6 });
7
8 server.listen(port, () => {
9   console.log(`Server running at port ${port}`);
10 });
11
```

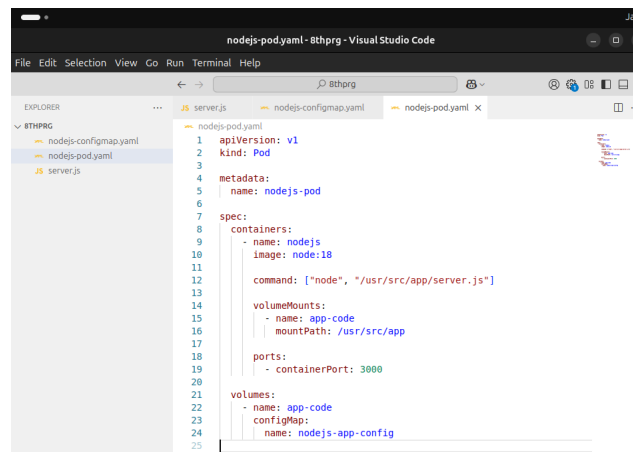
STEP 2: ConfigMap (`nodejs-configmap.yaml`)



The screenshot shows the Visual Studio Code editor with the `nodejs-configmap.yaml` file open. The code is as follows:

```
1 apiVersion: v1
2 kind: ConfigMap
3
4 metadata:
5   name: nodejs-app-config
6
7 data:
8   server.js: |
9     const http = require("http");
10
11     const server = http.createServer((req, res) => {
12       res.end("Hello from Node.js running inside Kubernetes!");
13     });
14
15     server.listen(3000, () => {
16       console.log("Server running on port 3000");
17     });
18
```

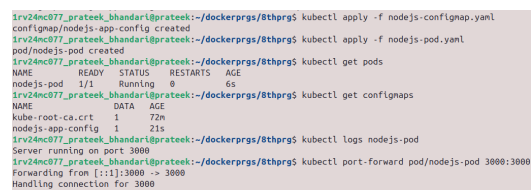
STEP 3: Pod YAML (`nodejs-pod.yaml`)

A screenshot of the Visual Studio Code editor interface. The Explorer sidebar on the left shows a project named 'BTHPRG' with files 'nodejs-configmap.yaml', 'nodejs-pod.yaml', and 'server.js'. The main editor window displays the 'nodejs-pod.yaml' file with the following content:

```
1 apiVersion: v1
2 kind: Pod
3
4 metadata:
5   name: nodejs-pod
6
7 spec:
8   containers:
9     - name: nodejs
10       image: node:18
11
12       command: ["node", "/usr/src/app/server.js"]
13
14       volumeMounts:
15         - name: app-code
16           mountPath: /usr/src/app
17
18       ports:
19         - containerPort: 3000
20
21   volumes:
22     - name: app-code
23       configMap:
24         name: nodejs-app-config
25
```

STEP 4: Apply the YAML files

STEP 5: Check pod status

A terminal window showing a series of Kubernetes commands and their outputs. The commands and outputs are:

```
1rv24nc077_prateek_bhandardi@prateek:~/dockerprgs/Bthprg$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config created
2rv24nc077_prateek_bhandardi@prateek:~/dockerprgs/Bthprg$ kubectl apply -f nodejs-pod.yaml
pod/nodejs-pod created
3rv24nc077_prateek_bhandardi@prateek:~/dockerprgs/Bthprg$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
nodejs-pod    1/1     Running   0           6s
4rv24nc077_prateek_bhandardi@prateek:~/dockerprgs/Bthprg$ kubectl get configmaps
NAME          DATA   AGE
nodejs-app-config 1       21s
5rv24nc077_prateek_bhandardi@prateek:~/dockerprgs/Bthprg$ kubectl logs nodejs-pod
Server running on port 3000
6rv24nc077_prateek_bhandardi@prateek:~/dockerprgs/Bthprg$ kubectl port-forward pod/nodejs-pod 3000:3000
Forwarding from [::]:3000 -> 3000
Handling connection for 3000
```

STEP 6 :Access the Node.js app

